ET timeseries in TRISHNA

How to compute a continuous ET timeseries?

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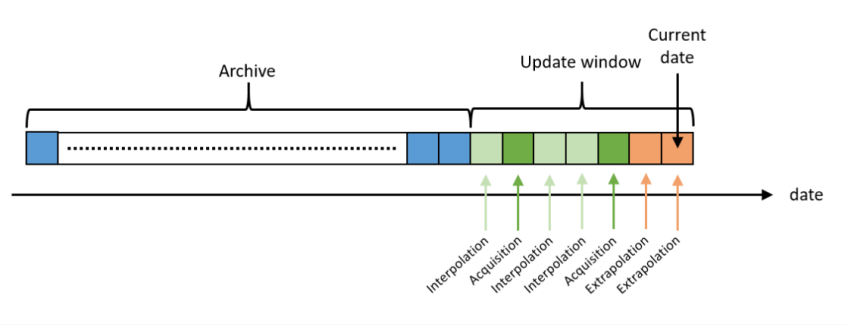
ET Timeseries

Overview

- Algorithm for calculating an evapotranspiration time series from L2B level products and auxiliary data

Algorithm

Evolution of a pixel's ET time series

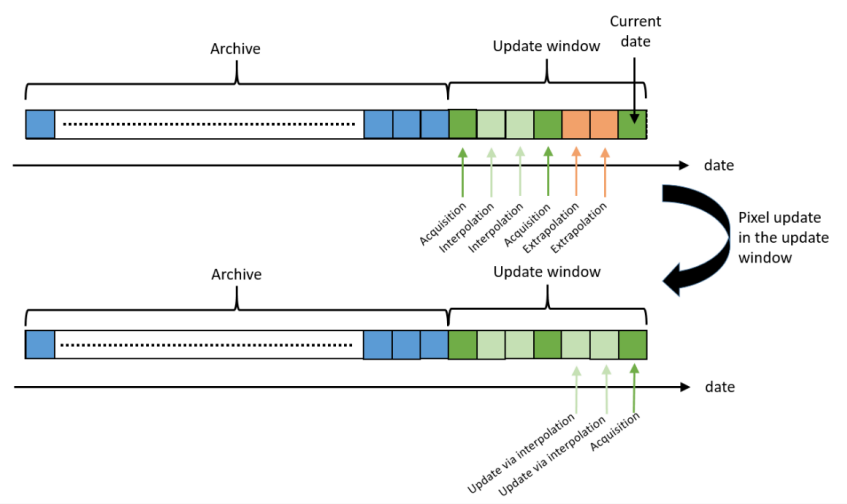


Algorithm

Update strategy

- interpolation between two existing acquisition values
- extrapolation beyond the last previous available acquisition value
- backward extrapolation from the next available acquisition value

Update timeseries



Algorithm

- V0: Implementation of interpolation/extrapolation based only a solar daily radiation

Algorithm step



Interpolation algorithm details

- interpolate the ratio of ET to an interpolation support

$$ET_d(t) = Ratio(t).B(t)$$

where B is the interpolation support

- use daily incoming solar radiation as interpolation support $\Rightarrow$ robust choice
  - *Solar radiation forms the major source of available energy, which explains the strong correlation with the EF and subsequently with dailyET.*

$$ET_d(t) = Ratio(t).R_{SD}(t)$$

Interpolation algorithm formula

- $Ratio(t)$ can be estimated using a simple linear interpolation between the available ratios corresponding to acquisitions $Ratio(t_k)$ and $Ratio(t_{k+1})$

$$Ratio(t) = Ratio\,(t_k)\ + (Ratio\,(t_{k+1}) -\ Ratio\,(t_k))\,\frac{t -\ t_k}{t_{k+1} - t_k}$$

Extrapolation algorithm formula

- The support B can be computed as corresponding to $Ratio(t_k)$ obtained on the day with the last available observation of ET

$$Ratio(t) \approx Ratio\,(t_k) = \frac{ET_d\,(t_k)}{B\,(t_k)}$$

L3 data preparation

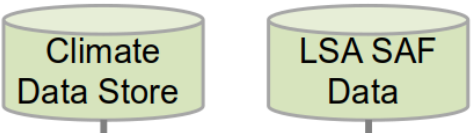
Input list

- Daily ET timeseries (used to simulation acquisition and also as reference)
- Daily radiation timeseries
- Other meteorological variables timeseries

Input source: Where the data is downloaded?

- Climate Data Store
- LSA SAF Data

Input source: What data is downloaded?



Configuration

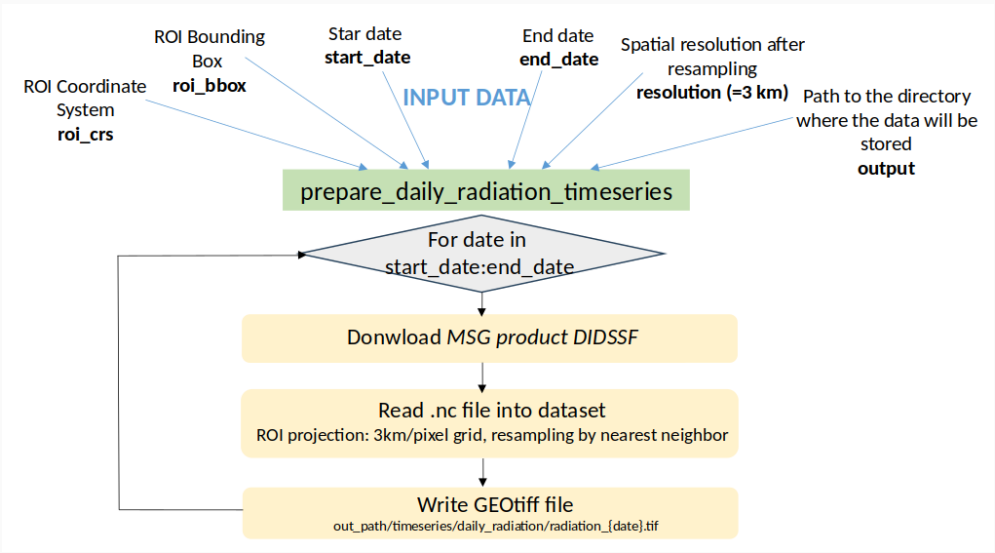
Climate Data Store

- Create an account
- Generate a token and store the token, see instructions

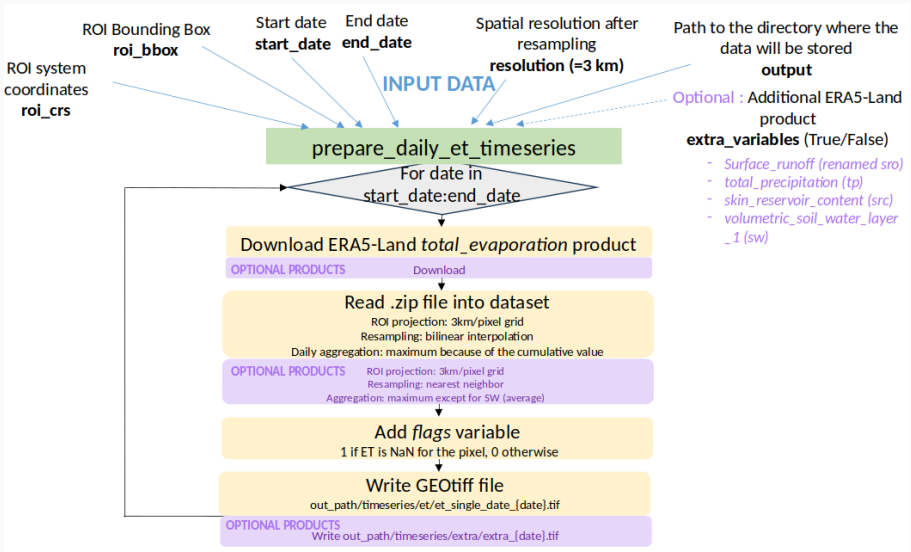
LSA SAF Data

- Create an account
- Set environment variables `LSASAF_USER` and `LSASAF_PASSWORD`

Prepare daily radiation timeseries



Prepare daily ET timeseries



Usage: Prepare timeseries script

Command line: scripts/prepare_timeseries.py

```
usage: prepare_timeseries.py [-h] -s START_DATE -e END_DATE -r ROI [-o OUTPUT]

options:
  -h, --help            show this help message and exit
  -s START_DATE, --start_date START_DATE
                        start date (YY-MM-DD)
  -e END_DATE, --end_date END_DATE
                        end date (YY-MM-DD)
  -r ROI, --roi ROI     Path of the region of interest in Shapefile format
  -o OUTPUT, --output OUTPUT
                        Output directory
```

Usage: Prepare timeseries script

Notebooks

- `notebooks/prepare_timeseries.ipynb`

Usage: Prepare timeseries script

Output directory

```
|-- ERA5_data
    `-- download_era5land_YYYY-MM-DD.zip
|-- MSG_data
|   |-- MSG_YYYY-MM-DD
|   |   `-- NETCDF4_LSASAF_MSG_DIDSSF_MSG-Disk_YYYYMMDD0000.nc
`-- timeseries
    |-- daily_radiation
    |   `-- radiation_YYYYMMDD.tif
    |-- et
    |   `-- et_single_date_YYYYMMDD.tif
     `-- extra
        `-- extra_YYYYMMDD.tif
```

ET timeseries in SWEAT

Usage: Command line

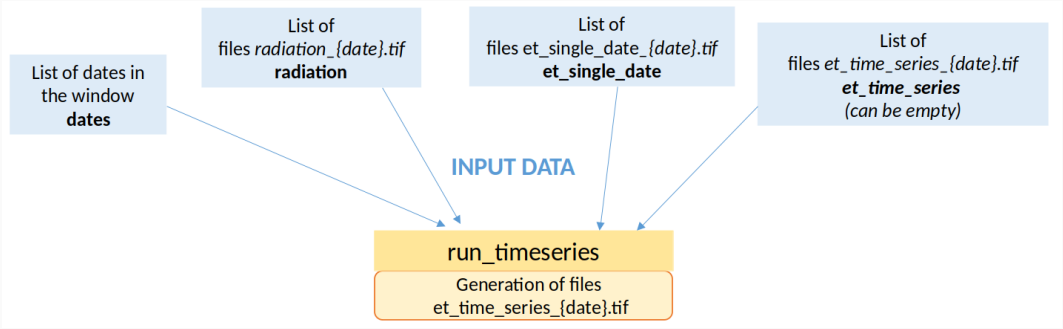
timeseries: Update ET timeseries over a window

```
Usage: timeseries [OPTIONS] INPUT_FILE

Options:
  --verbose / --no-verbose  Debug logging mode
  --version                 Show the version and exit.
  -h, --help                Show this message and exit.
```

Usage: Command line

timeseries: Update ET timeseries over a window



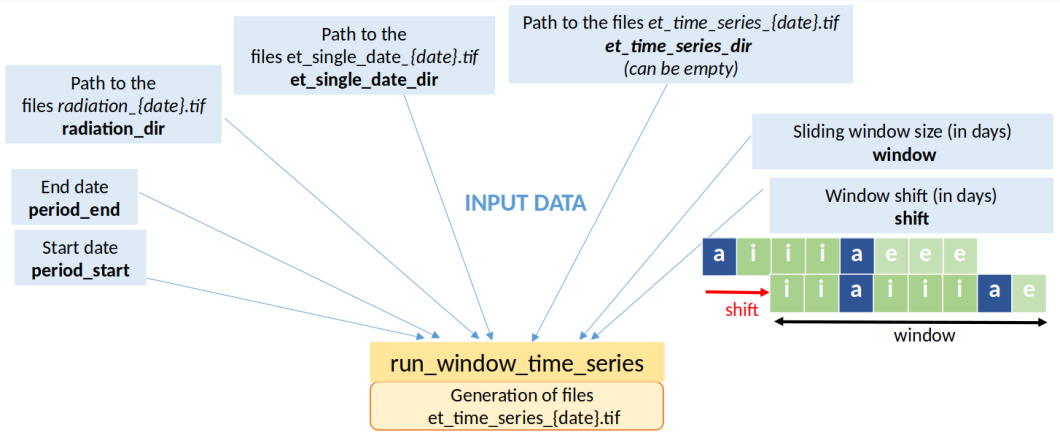
Usage: Command line

window_timeseries: Generate ET timeseries over a time period

```
Usage: window-timeseries [OPTIONS] INPUT_FILE

Options:
  --verbose / --no-verbose  Debug logging mode
  --version                 Show the version and exit.
  -h, --help                Show this message and exit.
```

window_timeseries: Generate ET timeseries over a time period



Usage: Notebook

Update ET timeseries over a window

- Main notebook **notebooks/run_timeseries.ipynb**

Generate ET timeseries over a time period

- Main notebook **notebooks/run_window_timeseries.ipynb**

Input configuration

- Detailed in usage page and configuration page

Exercises

- Test **tutorial_et_timeseries.ipynb**